

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
B.Tech Degree S4 (R,S) (FT/WP/PT) Exam April 2025 (2019 Scheme)

Course Code: CST 204

Course Name: Database Management Systems

Max. Marks: 100

Duration: 3 Hours

PART A

(Answer all questions; each question carries 3 marks)

Marks

- | | | |
|----|---|---|
| 1 | Compare weak entity and strong entity with examples. | 3 |
| 2 | What are the major differences between structured, unstructured and semi-structured data? | 3 |
| 3 | Compare SELECT and PROJECT operations on relational database with examples. | 3 |
| 4 | Differentiate between EQUI-JOIN, THETA JOIN and NATURAL JOIN | 3 |
| 5 | What is an assertion? How they differ from triggers? | 3 |
| 6 | Define primary key, candidate key and super key. | 3 |
| 7 | Define functional dependency and explain with example. | 3 |
| 8 | What are Armstrong's axioms? | 3 |
| 9 | What is the significance of Log-Based Recovery? | 3 |
| 10 | List and explain the desirable properties of a transaction | 3 |

PART B

(Answer one full question from each module, each question carries 14 marks)

Module -1

- | | | |
|----|--|----|
| 11 | a) With the help of a neat diagram explain the three schema architecture of DBMS | 7 |
| | b) What is an attribute? Explain different types of attributes in the data base management system. | 7 |
| 12 | a) Draw the ER model of a company by considering the following constraints | 10 |

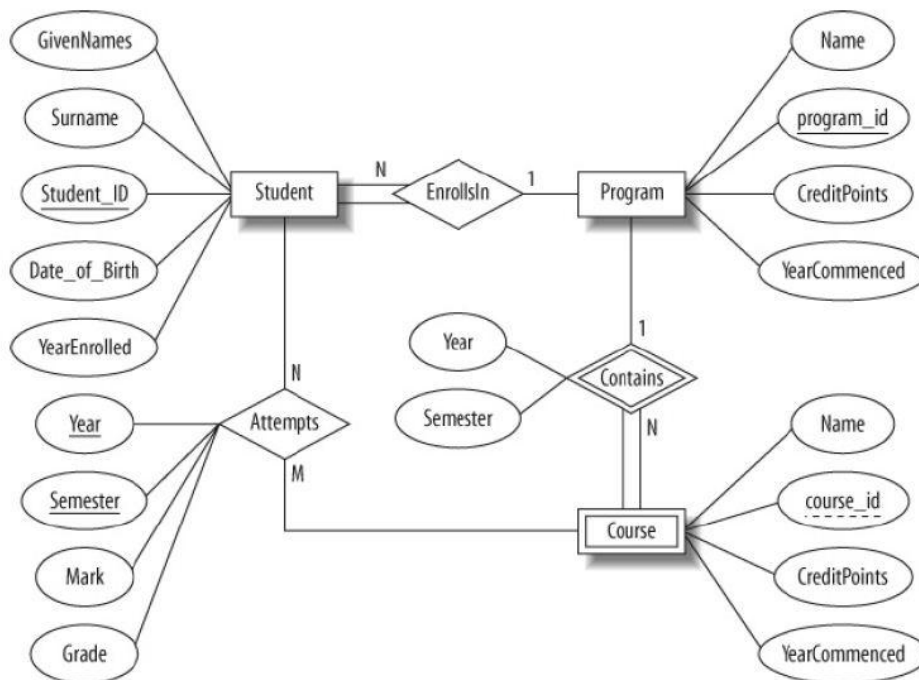
- In a company, an employee works on many projects which are controlled by one department.
- One employee supervises many employees.
- An employee has one or more dependents.
- One employee manages one department.

b) List different database users and explain each with examples.

4

Module -2

- 13 a) Write the rules for converting ER diagram to relational model and convert the following ER diagram to relational model. 10



b) Differentiate between the given below SQL statements with examples

4

- DROP and DELETE
- ALTER and UPDATE

- 14 a) An Employee relation has attributes: Employee-Id (numeric type), Name (character type), Salary (numeric type) and Dep-No (numeric type).
A Department relation has attributes: Department-Number (numeric type), Department-Name (character type), Dep-Manager-Id (numeric type).

Employee-Id is the primary key of Employee relation. Department-Number is the primary key of the Department relation. Dep-No attribute of Employee relation refers to the Department-Number attribute of Department relation and Dep-Manager-Id attribute of Department relation refers to the Employee-Id attribute of Employee relation.

(i) Write create table statements by specifying necessary integrity constraints for creating these two relations in SQL.

(ii) Write SQL statement to insert the details of an employee John with id 101 with salary 5000 and working in department number 5.

(iii) Insert the details of a Research Department with Department Number 1 and it has not been assigned any manager.

(iv) Assume that a department with employees working in it is to be deleted. Specify the two options to manage this scenario.

b) What is meant by constraints in relational database? Explain each with example. 6

Module -3

15 a) Illustrate structure of B-Tree and B+ Tree. Differentiate between internal nodes and leave nodes of both trees. 6

b) For the relation schema below, By assuming necessary key and referential integrity constraints, give an expression in SQL for each of the queries that follows: 8

employee (ID, person_name, street, city)

works (ID, company_name, salary)

company (company_name, city)

manages (ID, manager_id)

(i) Find the employees whose name starts with 'C'

(ii) Find the name of managers of each company

(iii) Find the ID, name, and city of residence of employees who works for “First Bank Corporation” and earns more than Rs. 50000

(iv) Find the name of companies whose employees earn a higher salary, on average, than the average salary at “First Bank Corporation”

- 16 a) Define views in SQL. Write simple SQL queries to create a view, update a view and drop a view. 8
- b) What is a grid file? What are its advantages and disadvantages? 6

Module -4

- 17 a) What is normalization? Explain about 1NF, 2NF and 3NF with definition and relevant examples. 7
- b) What are different anomalies in designing a database? Explain each with examples. 7
- 18 a) Consider the relation $R = \{A, B, C, D, E, F, G, H\}$ and the set of functional dependencies $F = \{A \rightarrow DE, B \rightarrow F, AB \rightarrow C, C \rightarrow GH, G \rightarrow H\}$. What is the key for R? Decompose R into 2NF and then 3NF relations. 8
- b) What is the dependency preservation property for decomposition? Why is it important? 6

Module -5

- 19 a) Differentiate serial and concurrent schedules. Elaborate conflict serializability with suitable example. 10
- b) Explain briefly the ACID properties of a transaction. 4
- 20 a) What are dirty-read and lost-update problems? Explain with the help of examples. 7
- b) What is the two-phase locking (2PL) protocol? How does it guarantee serializability? How strict 2PL differs from basic 2PL? 7

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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Fourth Semester B.Tech Degree (S, FE) Examination January 2024 (2019 Scheme)

Course Code: CST 204

Course Name: Database Management Systems

Max. Marks: 100

Duration: 3 Hours

PART A

(Answer all questions; each question carries 3 marks)

Marks

- | | | |
|----|--|---|
| 1 | Differentiate between Structured, Semi-structured, and Unstructured data. Give an example each. | 3 |
| 2 | Explain the three categories of Data Models. | 3 |
| 3 | Explain the "Cross-Reference approach" to mapping a Binary 1:1 Relationship Type that you employ when you map an ER Model into a Relational Schema, with the help of an example. | 3 |
| 4 | With the help of an example database, explain the usage of the set operations Union, Intersection, and set difference in Relational Algebra. | 3 |
| 5 | "Views simplify the specification of certain queries". Justify this statement | 3 |
| 6 | Differentiate between Assertions and Triggers. | 3 |
| 7 | Define the term "Functional Dependency". Give an example. | 3 |
| 8 | Give an algorithm to find the Minimal Cover for a set of Functional Dependencies. | 3 |
| 9 | Give six types of failures in a transaction processing system that necessitate Recovery procedures. | 3 |
| 10 | Explain the importance of transaction logging and checkpointing for efficient transaction processing. | 3 |

PART B

(Answer one full question from each module, each question carries 14 marks)

Module -1

- | | | |
|----|---|---|
| 11 | a) Explain the main four characteristics of the Database Approach that distinguish it from the traditional file-processing approach | 7 |
| | b) Categorize the different types of people who work in a database system environment. | 7 |

- 12 Design an ER diagram for a typical college library database and then map it into a relational database schema. List your assumptions and indicate the cardinality mappings. 14

Module -2

- 13 a) Consider the following schema and frame Relational Algebra queries for the following problems: 11

Suppliers (SID: integer, SName: string, Address: string) Parts (PID: integer,

PName: string, Color: string) Catalog (SID: integer, PID: integer, Cost: real)

The key fields are underlined and the domain of each field is given after the field name.

- (i) Find the names of suppliers who supply **red** parts
 - (ii) Find the SIDs of suppliers who supply some **red** part **or** are at the address '221 Packer Ave'
 - (iii) Find the SIDs of suppliers who supply some **red** part **and** some **green** part
 - (iv) Find the SIDs of suppliers who supply **every red** part
- b) Differentiate between DELETE and DROP commands in SQL. Illustrate their usage. 3

- 14 a) Consider a company database having the following schema and frame Relational Algebra queries for the following problems. Primary keys are underlined. 10

EMPLOYEE (SSN, Name, SupervisorSSN, Dnum)

DEPARTMENT (Dnumber, Dname, MgrSSN)

PROJECT (Pnumber, Plocation, ControlDeptNum)

EMPLOYEE (Dnum) References DEPARTMENT (Dnumber) and

PROJECT (ControlDeptNum) References DEPARTMENT (Dnumber)

- (i) Find the names of all employees who are supervised by the supervisor of the employee named 'Smith'
 - (ii) List the numbers of the projects (Pnumbers) controlled by Smith's department
- b) Illustrate the usage of the SQL commands – ALTER, INSERT, DELETE and UPDATE 4

Module -3

- 15 Consider the following Database with two tables: 14

Table: Employees

EmployeeID INT PRIMARY KEY

FirstName VARCHAR(50)

LastName VARCHAR(50)

JobTitle VARCHAR(100)

Salary DECIMAL(10,2)

HireDate DATE

DepartmentID INT

Table: Departments

DepartmentID INT PRIMARY KEY

DepartmentName VARCHAR(100)

ManagerID INT

Foreign Key: Employees.DepartmentID references Departments.DepartmentID

Frame SQL queries for the following problems:

- (i) Calculate the average salary per department.
 - (ii) List the employees with the highest salary in each department:
 - (iii) Find departments with more than 25 employees.
 - (iv) Get the employee names starting with 'S' in alphabetical order
- 16 a) With the help of an example explain Single-level indexing and multi-level indexing. Also, compare and contrast single-level indexing with multi-level indexing 8
- b) Explain a situation where a multi-level index would be significantly less effective than a single-level index, and vice versa 6

Module -4

- 17 a) Given the following FDs for the relation Employees (EmployeeID, DepartmentID, ManagerID, Salary): 6
- EmployeeID → DepartmentID
- DepartmentID → ManagerID

ManagerID $\rightarrow$ Salary

Identify any redundant FDs in the set and explain why they are redundant.

- b) Consider the following FDs for the relation Books (BookID, Title, Author, Publisher): 8

BookID $\rightarrow$ Title, Author

Author $\rightarrow$ Publisher

{Title, Publisher} $\rightarrow$ BookID

Find a minimal cover for this set of FDs. Explain how you arrived at your answer.

- 18 a) Give an algorithm each for checking Lossless Join and Dependency Preserving Properties 6

- b) Suppose that we decompose the schema $R = (A, B, C, D, E)$ into (A, B, C) (A, D, E). 8

Show that this decomposition is a lossless decomposition if the following set F of functional dependencies holds:

$F = \{A \rightarrow BC, CD \rightarrow E, B \rightarrow D, E \rightarrow A\}$

Module -5

- 19 a) What is a serial schedule? Why are serial schedules unacceptable in practice? 4
- b) What is a conflict serializable schedule? Give an algorithm to check whether a schedule is conflict serializable or not. Check whether the following schedules are conflict serializable or not and find an equivalent serial schedule if possible. 10
- (i) $r2(X); w2(X); r1(X); w1(X); r1(Y); w1(Y)$
- (ii) $r1(X); r2(X); w1(X); r1(Y); w2(X); w1(Y)$

- 20 a) Explain the working of Binary Locks and Shared/Exclusive Locks 5
- b) Explain Two-Phase locking protocol and any three variants of it 9

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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Fourth Semester B.Tech Degree Examination June 2022 (2019 scheme)

**Course Code: CST204****Course Name: DATABASE MANAGEMENT SYSTEMS**

Max. Marks: 100

Duration: 3 Hours

PART A*(Answer all questions; each question carries 3 marks)*

Marks

- | | | |
|----|--|---|
| 1 | List any three characteristics of database system | 3 |
| 2 | Draw neat labelled diagram of three schema architecture and briefly describe each level | 3 |
| 3 | Write briefly about any three relational database integrity constraints. | 3 |
| 4 | Differentiate between theta join and natural join operations. | 3 |
| 5 | Give any three uses of a trigger | 3 |
| 6 | A file has $r = 20000$ STUDENT records of fixed length. Each record has the following fields: NAME (30 bytes), SSN (9 bytes), ADDRESS (40 bytes), PHONE (9 bytes), BIRTHDATE (8 bytes), GENDER (1 byte), DEPTID (4 bytes), CLASSCODE (4 bytes), and PROGID (3 bytes). An additional byte is used as a deletion marker. The file is stored on the disk with block size $B = 512$ bytes. | 3 |
| | a) Calculate the record size R in bytes. | |
| | b) Calculate the blocking factor bfr and the number of file blocks b assuming an unspanned organization. | |
| | c) Calculate the average time it takes to find a record by doing a linear search | |
| 7 | Define Boyce-Codd normal form. How does it differ from 3NF? | 3 |
| 8 | Suppose, a relational schema $R(P, Q, R, S)$ and set of functional dependencies F and G are as follow: $F : \{ P \rightarrow Q, Q \rightarrow R, R \rightarrow S \}$ $G : \{ P \rightarrow QR, R \rightarrow S \}$. Check the equivalency of functional dependencies F and G . | 3 |
| 9 | Write briefly on log based recovery | 3 |
| 10 | Explain briefly the characteristics of Column family database. | 3 |

PART B*(Answer one full question from each module, each question carries 14 marks)***Module -1**

- | | | |
|----|---|---|
| 11 | a) Differentiate between two-tier and three-tier client-server database architecture with the help of neat labelled diagrams. | 7 |
|----|---|---|

b) Draw an ER diagram based on the following information,

7

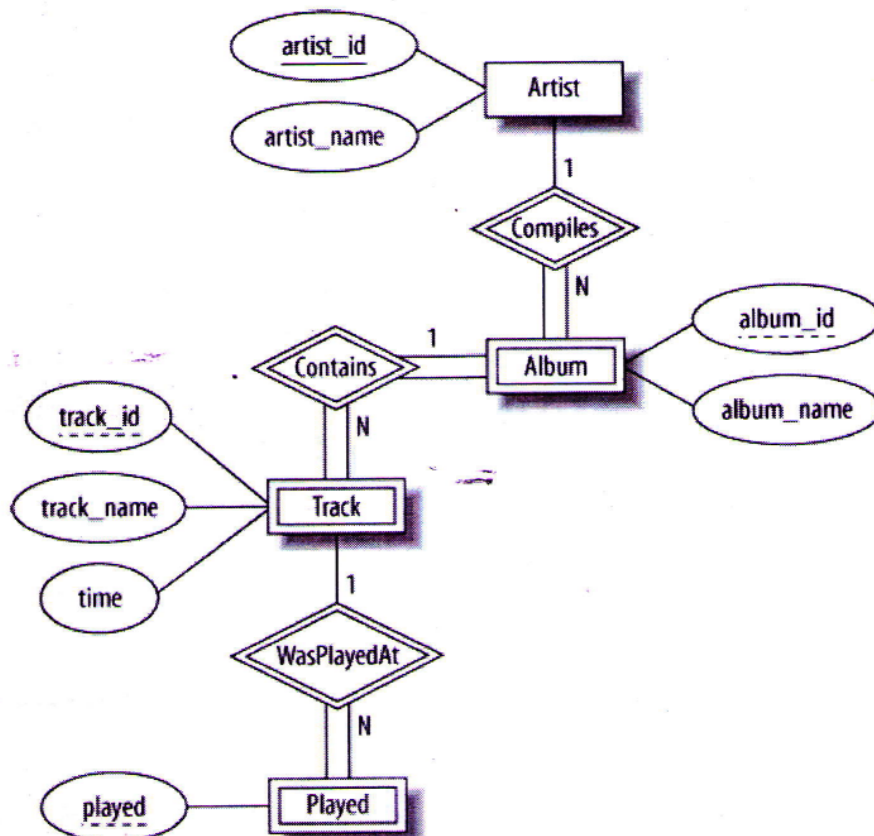
- Manufacturers have a name, which we may assume is unique, an address, and a phone number
- Products have a model number and a type. Each product is made by one manufacturer, and different manufacturers may have different products with the same model number. However, you may assume that no manufacturer would have two products with the same model number
- Customers are identified by their unique social security number. They have email addresses, and physical addresses. Several customers may live at the same (physical) address, but we assume that no two customers have the same email address
- An order has a unique order number, and a date. An order is placed by one customer. For each order, there are one or more products ordered, and there is a quantity for each product on the order.

12 a) Write briefly about any three types of database end users

6

b) Interpret the following ER diagram

8



Module -2

- 13 a) Consider the following schema, 8

Suppliers (sid, sname, address)

Parts (pid, pname, color)

Catalog (sid, pid, cost)

The primary key fields are underlined.

Write relational algebra expressions for the following queries:

- b) Find the name of parts supplied by supplier with sid=105
- ii) Find the names of suppliers supplying some green part for less than Rs 1000
- iii) Find the IDs of suppliers who supply some red or green part
- iv) Find the names of suppliers who supply some red part
- b) Differentiate between the following SQL statements 6
 - b) DROP and DELETE
 - ii) ALTER and UPDATE

- 14 a) Write SQL DDL statements based on the following database schema (Assume suitable domain types): 8

Employee (eid, name, designation, salary, comp_id)

Company (comp_id, cname, address, turnover)

- b) Create the above mentioned tables assuming each company has many employees. Mention the primary key, foreign key and not null constraints.
- ii) Insert values into both the tables. Mention in which order insertions will be carried out.
- ii) Modify the table Employee to include a new column "years_of_exp"
- iv) Increment the salary of employees whose salary is less than Rs25000 by 5%
- b) Illustrate any three ways of using INSERT statement in SQL. 6

Module -3

- 15 a) For the relation schema below, give an expression in SQL for each of the queries that follows: 8

employee (ID, person_name, street, city)

works (ID, company_name, salary)

company (company_name, city)

manages (ID, manager_id)

- b) Find the employees whose name starts with 'C'
- ii) Find the name of managers of each company
- iii) Find the ID, name, and city of residence of employees who works for "First Bank Corporation" and earns more than Rs50000
- iv) Find the name of companies whose employees earn a higher salary, on average, than the average salary at "First Bank Corporation"
- b) Differentiate correlated and non-correlated nested queries with suitable examples 6
- 16 a) What is multi-level indexing? How does it improve the efficiency of searching an index file? 8
- b) Insert the following keys, in the order given, into a B-tree of order 3: {10, 50, 20, 5, 22, 25} 6

Module -4

- 17 a) Consider a relation $R(A, B, C, D, E)$ with FDs $AB \rightarrow C, AC \rightarrow B, BC \rightarrow A, D \rightarrow E$. Determine all the keys of relation R . Also decompose the relation into collections of relations that are in BCNF. 8
- b) Write briefly on the different types of anomalies in designing a database. 6
- 18 a) Consider a relation schema $R(A, B, C, D)$ with the following functional dependencies $A \rightarrow B, B \rightarrow C, C \rightarrow D, D \rightarrow B$. Determine whether the decomposition of R into $R_1(A, B), R_2(B, C)$ and $R_3(B, D)$ is lossless or lossy. Write the complete steps. 6
- b) What is dependency preservation property for decomposition? Why is it important? 8

Module -5

- 19 a) Explain briefly the ACID properties of a transaction. 8
- b) Check whether the given schedules are conflict serializable or not 6
 - i) $S1 : R_1(X), R_2(X), R_1(Y), R_2(Y), R_3(Y), W_1(X), W_2(Y)$
 - ii) $S2 : R_1(X), R_2(X), R_2(Y), W_2(Y), R_1(Y), W_1(X)$
- 20 a) What is two phase locking protocol? How does it guarantee serializability? 8
- b) What are the main characteristics of NOSQL systems in the areas related to data models and query languages? 6

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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Fourth Semester B.Tech Degree Supplementary Examination June 2023 (2019 scheme)

Course Code: CST 204**Course Name: DATABASE MANAGEMENT SYSTEMS**

Max. Marks: 100

Duration: 3 Hours

PART A*(Answer all questions; each question carries 3 marks)*

Marks

- | | | |
|----|---|---|
| 1 | Classify the following cases into logical data independence and physical data independence.
(a) Creating an index for a data file
(b) Changing the integrity constraint
(c) Reorganizing the file | 3 |
| 2 | What is meant by a recursive relationship type? Give an example of recursive relationship type. | 3 |
| 3 | Consider the relational model constraints: domain constraint, key constraint, entity integrity and referential integrity. Specify which of these constraints may be violated during the following modification operations: insert, update and delete. | 3 |
| 4 | What is meant by complete set of relational algebra? Show how join operation in relational algebra can be represented using this set. | 3 |
| 5 | What is meant by a correlated nested query? Give a suitable example. | 3 |
| 6 | Explain the advantage of a multilevel index. | 3 |
| 7 | Why Armstrong's axioms are said to sound and complete? | 3 |
| 8 | What is meant by lossless join property? | 3 |
| 9 | List and explain the desirable properties of a transaction. | 3 |
| 10 | Illustrate the states for transaction execution. | 3 |

PART B*(Answer one full question from each module, each question carries 14 marks)***Module -1**

- | | | |
|----|---|----|
| 11 | Consider the following information about a university database: Professors have a ssn, a name, an age, a rank, and a research specialty. Projects have a project number, a sponsor name, a starting date, an ending date, and a budget. Graduate students have ssn, a name, an age, and a degree program (e.g., M.S. or Ph.D.). Each project is managed by one professor (known as the project's principal investigator). Each project is worked on by one or more professors (known as the project's co-investigators). Professors can manage and/or work on multiple projects. Each project is worked on by one or more graduate students (known as the project's research assistants). When graduate students work on a project, a professor must supervise their work on the project. Departments have a department number, a department name, and a location. Departments have a professor (known as the chairman) who manages the department. Professors work in one or more departments, and for each department that they work in, a time | 14 |
|----|---|----|

percentage is associated with their job. Graduate students have one major department in which they are working on their degree. Each graduate student has another, more senior graduate student (known as a student advisor) who advises him or her on what courses to take. Design and draw an ER diagram that captures the information about the university.

- 12 a) Explain the difference between database schema and database state with suitable example. 7
Specify the role of schema in a DBMS.
- b) With a neat diagram explain the Three Schema Architecture of a DBMS 7

Module -2

- 13 a) Consider the two relations T1 and T2 shown below. Show the results of the following operations. 8

Relation T1

P	Q	R
30	Ac	25
35	Bc	28
45	Ac	26

Relation T2

A	B	C
30	Bc	26
45	Cc	23
30	Bc	25

- i) $T1 \bowtie_{T1.Q=T2.B} T2$
- ii) $T1 \bowtie_{T1.P=T2.A} T2$
- iii) $T1 \cup T2$
- iv) $T1 \bowtie_{(T1.P=T2.A \text{ AND } T1.R=T2.C)} T2$

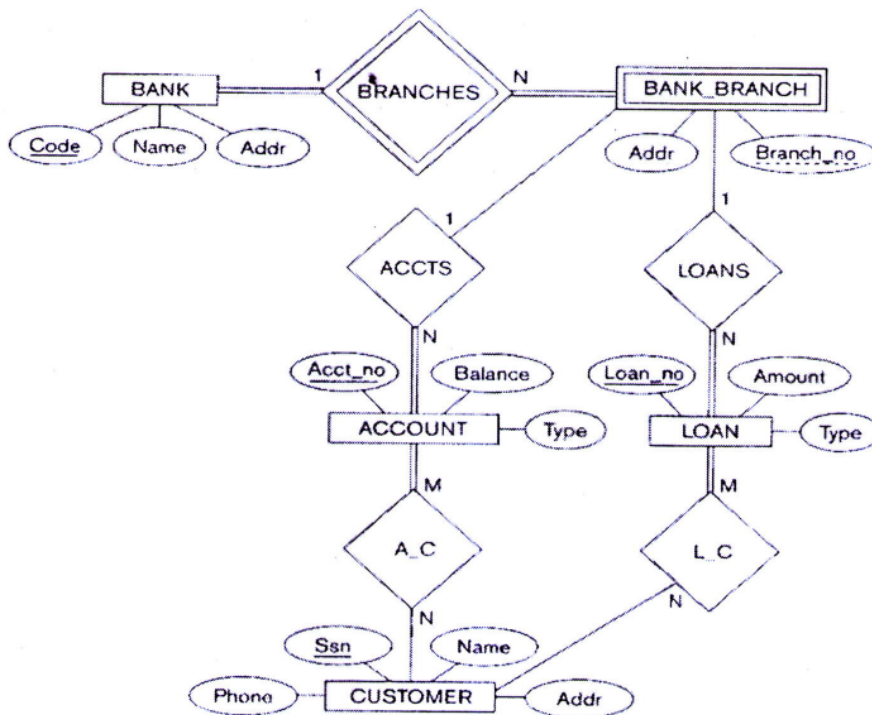
- b) An Employee relation has attributes: Employee-Id (numeric type), Name (character type), Salary (numeric type) and Dep-No (numeric type). 6

A Department relation has attributes: Department-Number (numeric type), Department-Name(character type), Dep-Manager-Id (numeric type).

Employee-Id is the primary key of Employee relation. Department-Number is the primary key of the Department relation. Dep-No attribute of Employee relation refers to the Department-Number attribute of Department relation and Dep-Manager-Id attribute of Department relation refers to the Employee-Id attribute of Employee relation.

- (i) Write create table statements by specifying necessary integrity constraints for creating these two relations in SQL.
- (ii) Write SQL statement to insert the details of an employee John with id 101 with salary 5000 and working in department number 5.
- (iii) Insert the details of a Research Department with Department Number 1 and it has not been assigned any manager.
- (iv) Assume that a department with employees working in it is to be deleted. Specify the two options to manage this scenario.

- 14 Convert the ER schema for Bank database given below into a relational schema. Specify all primary keys and foreign keys. 14



Module -3

- 15 a) Consider the following relations: 7
 Employee (Employee-Id, Employee-Name, Salary, Department-No)
 Department (Department-No, Department-Name)
 Write SQL queries for the following:
- Retrieve the employee names and their department names
 - Retrieve department names and the average salary given by them
 - Retrieve the ids of employees getting salary greater than the average salary of their department
 - For each department that has more than 4 employees, retrieve the department-No and the number of employees getting salary more than Rs. 50000
- b) What is meant by a heap file? Explain how insert, update, delete and search operations can be performed in a heap file. 7
- 16 a) What are the advantages of Views? Explain two view implementation techniques. 7
- b) Consider a disk with block size 512 bytes. A block pointer is 6 bytes long, and a record pointer is 7 bytes long. A file has 30,000 EMPLOYEE records of fixed-length. Each record has the following fields: NAME (30 bytes), SSN (9bytes), DEPARTMENTCODE (9 bytes), ADDRESS (40 bytes), PHONE (9 bytes), BIRTHDATE (8 bytes), SEX (1 byte), JOBCODE (4 bytes), SALARY (4 bytes, real number). An additional byte is used as a deletion marker. Assume that file is not ordered by the key field SSN and we need to create a secondary index on SSN. 7
- Find the number of levels needed, if we make it into a multilevel index.
 - Find the number of block accesses needed to retrieve a record from this file if we use the multilevel index.

Module -4

- 17 a) Given relation $R(A,B,C,D,E)$ and functional dependencies $F=\{AB \rightarrow C, CE \rightarrow D, A \rightarrow E\}$. Determine whether each functional dependency below is in F^+ or not: 6
- $AB \rightarrow D$
 - $A \rightarrow C$
- b) Consider the following relation: 8
- $CAR_SALE(Car\#, Date_sold, Salesperson\#, Commission\%, Discount_amt)$
- Assume that a car may be sold by multiple salespeople, and hence $\{Car\#, Salesperson\# \}$ is the primary key. Additional dependencies are :
- $Date_sold \rightarrow Discount_amt$ and $Salesperson\# \rightarrow Commission\%$
- Based on the given primary key and functional dependencies, is this relation in 1NF, 2NF, or 3NF? Why or why not?
 - How would you successively normalize it completely?
- 18 a) Consider the following decompositions for the relation schema R into R_1, R_2 and R_3 . Determine whether the decomposition has the lossless join property with respect to the given F . 7
- $R=\{P, Q, R, S, T, U\}$
- $R_1=\{P, Q\}, R_2=\{R, S, T\}, R_3=\{P,R,U\}$
- $F = \{ P \rightarrow Q, R \rightarrow \{S, T\}, \{P,R\} \rightarrow U \}$
- b) Explain insert, update and delete anomalies with suitable examples. 7

Module -5

- 19 a) Consider the schedule S of three transactions T_1, T_2 and T_3 given below. State whether the schedule is serializable or not. 7
- $S: r_3(Y), r_3(Z), r_1(X), w_1(X), w_3(Y), w_3(Z), r_2(Z), r_1(Y), w_1(Y), r_2(Y), w_2(Y), r_2(X), w_2(X)$
- (Hint: Interpret the notation $r_3(Y)$ as the operation read database item Y of transaction T_3 .)
- b) Explain the lost update problem and temporary update problem that occur when concurrent execution is uncontrolled. 7
- 20 a) Explain conservative and strict two-phase locking techniques. Why strict 2PL is deadlock free? 7
- b) Differentiate among recoverable, cascading rollback and strict schedules with suitable examples. 7

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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S4 (R,S) / S4 (PT) (R,S) / S4 (WP) (R) Examination May 2024 (2019 Scheme)

**Course Code: CST204****Course Name: Database Management Systems****Max. Marks: 100****Duration: 3 Hours****PART A***(Answer all questions; each question carries 3 marks)***Marks**

- | | | |
|----|---|---|
| 1 | List out any three database users and their functionalities. | 3 |
| 2 | Distinguish between schema and instance. | 3 |
| 3 | Explain the intersection operation in relational algebra with an example. | 3 |
| 4 | Differentiate between Super key, Candidate key and Primary key. | 3 |
| 5 | Outline concept of assertion in SQL with an example. | 3 |
| 6 | Explain any three differences between Hash indexes and B+ tree indexes. | 3 |
| 7 | State the Armstrong's axioms of FD. | 3 |
| 8 | State insertion anomaly with suitable example. | 3 |
| 9 | Explain the different properties of a transaction. | 3 |
| 10 | Summarize the purpose of using a lock compatibility matrix. | 3 |

PART B*(Answer one full question from each module, each question carries 14 marks)***Module -1**

- | | | |
|----|---|---|
| 11 | a) Explain the seven drawbacks of file processing system, compared to database approach. | 7 |
| | b) Represent a "Library Management Software" using an ER diagram. | 7 |
| 12 | a) Explain the concept of three schema architecture with the help of a neat labelled diagram. | 7 |
| | b) Represent an "Online Shopping Portal" using an ER diagram. | 7 |

Module -2

- | | | |
|----|--|---|
| 13 | a) Explain the left outer join, right outer join and full outer join operations with | 6 |
|----|--|---|

suitable examples.

- b) Consider the following schema.

Suppliers(sid, sname, address)

Parts(pid, pname, color)

Catalog(sid, pid, cost)

The primary key fields are underlined; Foreign keys have the same name as primary keys. Assume integer domain for the attributes sid, pid, cost and string domain for the attributes sname, address, pname, color. Write **relational algebra** expressions for the given questions. (Use * symbol for natural join and $\bowtie$ symbol for join)

- (i) Find the names of suppliers who supply some red part.
- (ii) Find the sids of suppliers who supply some red or green part.
- (iii) Find the sids of suppliers who supply every red part or supply every green part.
- (iv) Find the pids of parts supplied by at least two different suppliers.

- 14 a) What is constraint? Discuss about domain constraint, entity integrity and referential integrity constraint with suitable example. 7
- b) Illustrate with an example the different steps involved in synthesizing an ER diagram into a relational schema. 7

Module -3

- 15 a) With the help of an example explain Single-level indexing and multi-level indexing. Explain the difference (any four) between single-level indexing and multi-level indexing. 8
- b) Describe any three aggregate functions in SQL with example. 6
- 16 a) Explain a situation where a multi-level index would be significantly less effective than a single-level index, and vice versa. 6
- b) Consider the following relations: 8

Employee (Employee-Id, Employee-Name, Salary, Department-No)

Department (Department-No, Department-Name)

The primary key fields are underlined. Foreign keys have the same name as primary keys. Write **SQL** queries for the following:

- (i) Retrieve the employee names and their department names.
- (ii) Retrieve department names and the average salary given by them.

- (iii) Retrieve the ids of employees getting salary greater than the average salary of their department.
- (iv) For each department that has more than 4 employees, retrieve the department-No and the number of employees getting salary more than Rs. 50000.

Module -4

- 17 a) Write an algorithm to find the closure of an attribute. 6
- b) Let $R = (A, B, C, D, E)$, $R_1 = (A, D)$, $R_2 = (A, B)$, $R_3 = (B, E)$, $R_4 = (C, D, E)$, and $R_5 = (A, E)$. Let the FDs be: $A \rightarrow C$, $B \rightarrow C$, $C \rightarrow D$, $A \rightarrow D$, $DE \rightarrow C$, $CE \rightarrow A$. Check whether the decomposition of R into $\{R_1, R_2, R_3, R_4, R_5\}$ is a lossless decomposition or not. 8
- 18 a) P and Q are two set of FDs for a relational schema $R(A, B, C, D)$. $P = \{A \rightarrow B, B \rightarrow C, C \rightarrow D\}$ and $Q = \{A \rightarrow BC, C \rightarrow D\}$. Check whether P covers Q and Q covers P ? Also check whether P and Q are equivalent? 6
- b) Define 1NF, 2NF, 3NF and BCNF with suitable examples. 8

Module -5

- 19 a) Describe the different states of a transaction with the help of a neat sketch. 7
- b) What is deferred database modification? How it is different from immediate database modification? Explain the recovery steps in deferred database modification with an example. 7
- 20 a) Determine if the following schedule is recoverable. Is the schedule cascadeless? Justify your answer. 4
- $r_1(X), r_2(Z), r_1(Z), r_3(X), r_3(Y), w_1(X), c_1, w_3(Y), c_3, r_2(Y), w_2(Z), w_2(Y), c_2$
(Note: $ri(X)/wi(X)$ means transaction T_i issues read/write on item X ; ci means transaction T_i commits.)
- b) What is REDIS? What are the Features (any four) of REDIS? 6
- c) Check whether the following schedule is conflict serializable or not and find an equivalent serial schedule if possible. 4
- $r_1(X), r_2(Z), r_1(Z), r_3(X), r_3(Y), w_1(X), w_3(Y), r_2(Y), w_2(Z), w_2(Y)$
(Note: $ri(X)/wi(X)$ means transaction T_i issues read/write on item X)

Sample Course Level Assessment Questions

Course Outcome1 (CO1):

1. List out any three salient features of database systems, which distinguish it from a file system.
2. Give one example each for logical and physical data independence.

Course Outcome 2(CO2):

1. What facts about the relationships between entities EMPLOYEE and PROJECT are conveyed by the following ER diagram?



1. Design an ER diagram for the following scenario:
There is a set of teams, each team has an ID (unique identifier), name, main stadium, and to which city this team belongs. Each team has many players, and each player belongs to one team. Each player has a number (unique identifier), name, DoB, start year, and shirt number that he uses. Teams play matches, in each match there is a host team and a guest team.

Course Outcome 3(CO3):

1. For the SQL query, `SELECT A, B FROM R WHERE B = 'apple' AND C = 'orange'` on the table `R(A, B, C, D)`, where A is a key, write any three equivalent relational algebra expressions.
2. Given the FDs $P \rightarrow Q$, $P \rightarrow R$, $QR \rightarrow S$, $Q \rightarrow T$, $QR \rightarrow U$, $PR \rightarrow U$, write the sequence of *Armstrong's Axioms* needed to arrive at the following FDs: (a) $P \rightarrow T$ (b) $PR \rightarrow S$ (c) $QR \rightarrow SU$
3. Consider a relation PLAYER (PLAYER-NO, PLAYER-NAME, PLAYER-POSN, TEAM, TEAM-COLOR, COACH-NO, COACH-NAME, TEAM-CAPTAIN). Assume that PLAYER-NO is the *only* key of the relation and that the following dependencies hold:
 $TEAM \rightarrow \{TEAM-COLOR, COACH-NO, TEAM-CAPTAIN\}$
 $COACH-NO \rightarrow COACH-NAME$
 - i. Is the relation in 2NF? If not, decompose to 2NF.
 - ii. Is the relation in 3NF? If not, decompose to 3NF.

4. In the following tables foreign keys have the same name as primary keys except DIRECTED-BY, which refers to the primary key ARTIST-ID. Consider only *single-director* movies.

MOVIES(MOVIE-ID, MNAME, GENRE, LENGTH, DIRECTED-BY)

ARTIST(ARTIST-ID, ANAME)

ACTING(ARTIST-ID, MOVIE-ID)

Write SQL expressions for the following queries:

- (a) Name(s) and director name(s) of movie(s) acted by 'Jenny'.
- (b) Names of actors who have never acted with 'Rony'
- (c) Count of movies genre-wise.
- (d) Name(s) of movies with maximum length.

Course Outcome 4(CO4):

1. Consider an EMPLOYEE file with 10000 records where each record is of size 80 bytes. The file is sorted on employee number (15 bytes long), which is the primary key. Assuming un-spanned organization, block size of 512 bytes and block pointer size of 5 bytes. Compute the number of block accesses needed for retrieving an employee record based on employee number if (i) No index is used (ii) Multi-level primary index is used.

Course Outcome 5(CO5):

1. Determine if the following schedule is *recoverable*. Is the schedule *cascade-less*? Justify your answer. $r1(X), r2(Z), r1(Z), r3(X), r3(Y), w1(X), c1, w3(Y), c3, r2(Y), w2(Z), w2(Y), c2$. (Note: $ri(X)/wi(X)$ means transaction T_i issues read/write on item X; ci means transaction T_i commits.)
2. Two-phase locking protocol ensures serializability. Justify.

Course Outcome 6(CO6):

1. List out any three salient features of NoSQL databases. Give example of a document in MongoDB.

Model Question paper

QPCODE

Reg No: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

FOURTH SEMESTER B.TECH DEGREE EXAMINATION, MONTH & YEAR

Course Code: CST 204

Course Name: Database Management Systems

Max.Marks:100

Duration: 3 Hours

PART A

Answer all Questions. Each question carries 3 Marks

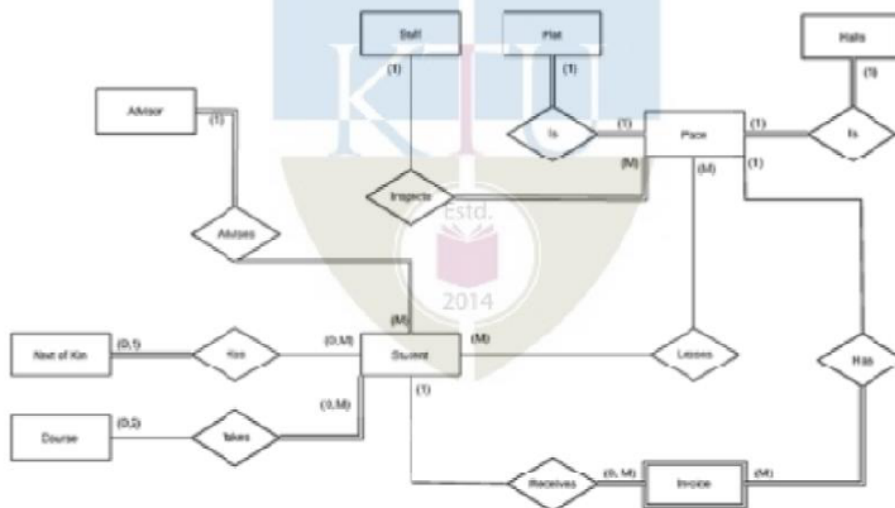
- 1 List out any three salient features of a database systems.
- 2 When is multi-valued composite attribute used in ER modelling?
- 3 For the SQL query, *SELECT A, B FROM R WHERE B='apple' AND C = 'orange'* on the table R(A, B, C, D), where A is a key, write any two equivalent relational algebra expressions.
- 4 Outline the concept of *theta*-join.
- 5 How is the purpose of *where* clause is different from that of having clause?
- 6 What is the use of a trigger?
- 7 When do you say that a relation is not in 1NF?
- 8 Given the FDs $P \rightarrow Q$, $P \rightarrow R$, $QR \rightarrow S$, $Q \rightarrow T$, $QR \rightarrow U$, $PR \rightarrow U$, write the sequence of Armstrong's Axioms needed to arrive at a. $P \rightarrow T$ b. $PR \rightarrow S$
- 9 What is meant by the lost update problem?
- 10 What is meant by check pointing?

Answer any one Question from each module. Each question carries 14 Marks

- 11 a. Design an ER diagram for the following scenario: There is a set of teams, each team has an ID (unique identifier), name, main stadium, and to which city this team belongs. Each team has many players, and each player belongs to one team. Each player has a number (unique identifier), name, DoB, start year, and shirt number that he uses. Teams play matches, in each match there is a host team and a guest team. The match takes place in the stadium of the host team. For each match we need to keep track of the following: The date on which the game is played The final result of the match. The players participated in the match. For each player, how many goals he scored, whether or not he took yellow card, and whether or not he took red card. During the match, one player may substitute another player. We want to capture this substitution and the time at which it took place. Each match has exactly three referees. For each referee we have an ID (unique identifier), name, DoB, years of experience. One referee is the main referee and the other two are assistant referee. (14)

OR

- 12 a. Interpret the the following ER diagram. (8)



- b. Distinguish between physical data independence and logical data independence with suitable examples. (6)

- 13 **EMPLOYEE**(ENO, NAME, ADDRESS, DOB, AGE, GENDER, SALARY, DNUM, SUPERENO) (14)
DEPARTMENT(DNO, DNAME, DLOCATION, DPHONE, MGRENO)
PROJECT(PNO, PNAME, PLOCATION, PCOST, CDNO)

DNUM is a foreign key that identifies the department to which an employee belongs. MGRENO is a foreign key identifying the employee who manages the department. CDNO is a foreign key identifying the department that controls the project. SUPERENO is a foreign key identifying the supervisor of each employee.

Write relational algebra expressions for the following queries:-

- (a) Names of female employees whose salary is more than 20000.
- (b) Salaries of employee from 'Accounts' department
- (c) Names of employees along with his/her supervisor's name
- (d) For each employee return name of the employee along with his department name and the names of projects in which he/she works
- (e) Names of employees working in all the departments

OR

- 14 a. Write SQL DDL statements for the the following (Assume suitable domain types): (10)
- i. Create the tables STUDENT(ROLLNO, NAME, CLASS, SEM, ADVISER), FACULTY(FID, NAME, SALARY, DEPT). Assume that ADVISER is a foreign key referring FACUTY table.
 - ii. Delete department with name 'CS' and all employees of the department.
 - iii. Increment salary of every faculty by 10%.
- b. Illustrate foreign key constraint with a typical example. (4)

- 15 For the relation schema below, give an expression in SQL for each of the queries (14) that follows:

employee(employee-name, street, city)
works(employee-name, company-name, salary)
company(company-name, city)
manages(employee-name, manager-name)

- Find the names, street address, and cities of residence for all employees who work for the Company 'RIL Inc.' and earn more than \$10,000.
- Find the names of all employees who live in the same cities as the companies for which they work.
- Find the names of all employees who do not work for 'KYS Inc.'. Assume that all people work for exactly one company.
- Find the names of all employees who earn more than every employee of 'SB Corporation'. Assume that all people work for at most one company.
- List out number of employees company-wise in the decreasing order of number of employees.

OR

- 16 a. Consider an EMPLOYEE file with 10000 records where each record is of size 80 bytes. The file is sorted on employee number (15 bytes long), which is the primary key. Assuming un-spanned organization and block size of 512 bytes compute the number of block accesses needed for selecting records based on employee number if,
- No index is used
 - Single level primary index is used
 - Multi-level primary index is used
- Assume a block pointer size of 6 bytes. (9)
- b. Illustrate correlated and non-correlated nested queries with real examples. (5)
- 17 a. Illustrate 3NF and BCNF with suitable real examples. (6)
- b. Given a relation $R(A_1, A_2, A_3, A_4, A_5)$ with functional dependencies $A_1 \rightarrow A_2 A_4$ and $A_4 \rightarrow A_5$, check if the decomposition $R_1(A_1, A_2, A_3)$, $R_2(A_1, A_4)$, $R_3(A_2, A_4, A_5)$ is lossless. (8)

OR

- 18 a. Consider the un-normalized relation $R(A, B, C, D, E, F, G)$ with the FDs $A \rightarrow B$, $AC \rightarrow G$, $AD \rightarrow EF$, $EF \rightarrow G$, $CDE \rightarrow AB$. Trace the normalization process to reach 3NF relations. (7)

- b. Illustrate Lossless Join Decomposition and Dependency Preserving Decomposition with typical examples. (7)
- 19 a. Discuss the four ACID properties and their importance. (7)
- b. Determine if the following schedule is conflict serializable. Is the schedule recoverable? Is the schedule cascade-less? Justify your answers. (7)
- $r1(X), r2(Z), r1(Z), r3(X), r3(Y), w1(X), c1, w3(Y), c3, r2(Y), w2(Z), w2(Y), c2$
- (Note: $ri(X)/wi(X)$ means transaction T_i issues read/write on item X ; ci means transaction T_i commits.)

OR

- 20 a. Discuss the main characteristics of Key-value DB and Graph DB. (7)
- b. Illustrate two-phase locking with a schedule containing three transactions. Argue that 2PL ensures serializability. Also argue that 2PL can lead to deadlock. (7)

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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S4 (S,FE) (FT/WP/PT) Examination December 2025/January 2026 (2019 Scheme)

Course Code: CST204**Course Name: Database Management Systems**

Max. Marks: 100

Duration: 3 Hours

PART A*(Answer all questions; each question carries 3 marks)*

Marks

- 1 Differentiate database schema and instance with suitable example. 3
- 2 Define the term: Composite attribute, Derived attribute, Multivalued attribute. 3
- 3 Discuss the entity integrity and referential integrity constraints. 3
- 4 Explain the difference between theta join and natural join with an example. 3
- 5 Write short note on views in SQL. 3
- 6 Differentiate sparse index and dense index with a neat diagram. 3
- 7 Identify which of the following functional dependencies may hold in the relation. 3
Specify the reason, if the dependency cannot hold.

 $XY \rightarrow Z$ $YZ \rightarrow X$ $XZ \rightarrow Y$

X	Y	Z
1	4	2
1	5	3
1	6	3
3	2	2

- 8 Explain the need of normalization in database design. 3
- 9 Explain the desirable properties of transaction. 3
- 10 List out any three salient features of NoSQL database. 3

PART B*(Answer one full question from each module, each question carries 14 marks)***Module -1**

- 11 a) Explain the advantage of using DBMS Approach. 8
b) Explain how database management systems are classified. 6

- 12 a) Draw an ER diagram based on the following information 8

Consider a MOVIE database in which data is recorded about the movie industry. The data requirements are summarized as follows:

Each movie is identified by title and year of release. Each movie has a length in minutes. Each has a production company, and each is classified under one or more genres (such as horror, action, drama, and so forth). Each movie has one or more directors and one or more actors appear in it. Each movie also has a plot outline. Finally, each movie has zero or more quotable quotes, each of which is spoken by a particular actor appearing in the movie.

- Actors are identified by name and date of birth and appear in one or more movies. Each actor has a role in the movie.
- Directors are also identified by name and date of birth and direct one or more movies. It is possible for a director to act in a movie (including one that he or she may also direct).
- Production companies are identified by name, and each has an address. A production company produces one or more movies.

- b) Explain the constraint on binary relationship types. 6

Module -2

- 13 a) Consider the following relational database schema consisting of the four relation schemas: 8

Passenger (pid, pname, pgender, pcity)

Agency (aid, aname , acity)

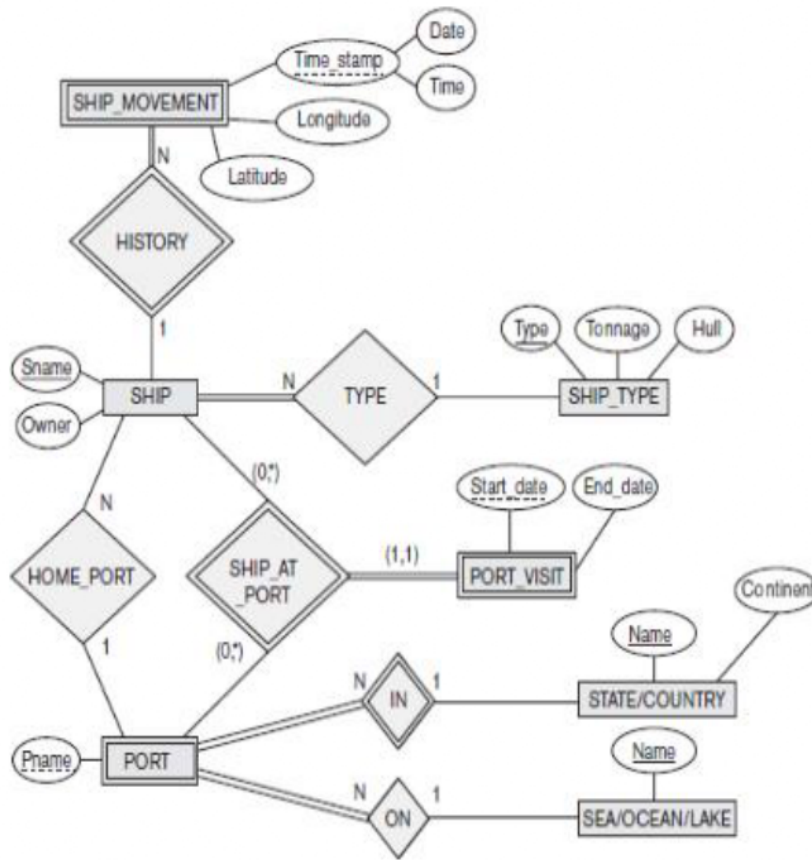
Flight (fid, fdate, time , src, dest)

Booking (pid, aid , fid , fdate)

Answer the following question using relational algebra queries.

- i) Find the passenger names for passengers who have bookings on at least one flight.
- ii) Find the passenger names for those who do not have any bookings in any flights.
- iii) Find the agency names for agencies that located in the same city as passenger with passenger id 123.
- iv) Find the details of all male passengers who are associated with Jet agency.

- b) Explain outer join with suitable example. 6
- 14 a) Convert the ER Schema of ship tracking database into a relational schema. Specify all primary keys and foreign keys. 8



- b) Differentiate alter, update and drop, delete statements in SQL 6

Module -3

- 15 a) Consider the relation schema 8

Flights (flno: integer, from: string, to: string, distance: integer, departs: time, arrives: time, price: real)

Aircraft (aid: integer, aname: string, cruising range: integer)

Certified (eid: integer, aid: integer)

Employees (eid: integer, ename: string, salary: integer)

Note that the Employees relation describes pilots and other kinds of employees as well; every pilot is certified for some aircraft, and only pilots are certified to fly.

Write SQL query for the following.

- i) Find the number of flights in the route from Los Angeles to Chicago.

- ii) For each pilot who is certified for more than three aircraft, find the eid and the maximum cruising range of the aircraft for which she or he is certified.
 - iii) Find the names of pilots whose salary is less than the price of the cheapest route from Los Angeles to Chicago.
 - iv) Find the names of pilots certified for some Boeing aircraft.
- b) Explain correlated and non-correlated queries 6
- 16 a) Consider an ordered file with $r = 300,00$ records stored on a disk with block size $B = 4096$ bytes. File records are of fixed size and are unspanned, with record length $r = 100$ bytes. Find the blocking factor for the file. Find the binary search time to access a data. 8
- Suppose the ordering key field of the file, $v = 9$ bytes long and a block pointer of size 6 bytes long. Find the binary search time to access a data after indexing is used. Find the number of levels and block access needed if multilevel indexing is used.
- b) Compare single level index and multi-level index. 6

Module -4

- 17 a) Compute the closure of the following set F of functional dependencies for the relation schema $R = (A, B, C, D, E)$. $A \rightarrow BC$, $CD \rightarrow E$, $B \rightarrow D$, $E \rightarrow A$. List the candidate keys for R . 8
- b) Explain different normal forms in database. 6
- 18 a) Consider a relation schema $R(A, B, C, D, E, F, G, H)$ with the following functional dependencies $AB \rightarrow CE$, $C \rightarrow DE$, $F \rightarrow GH$, $E \rightarrow F$. Determine whether the decomposition of R into $R_1(A, B)$, $R_2(B, C, D)$, $R_3(D, E, F)$ and $R_4(C, G, H)$ is lossless or lossy. Write the complete steps. 8
- b) Show that the following two sets of Functional dependency are equivalent. 6
- $F = \{ A \rightarrow C, AC \rightarrow D, E \rightarrow AD, E \rightarrow H \}$ and $G = \{ A \rightarrow CD, E \rightarrow AH \}$

Module -5

- 19 a) Explain the different variation of two phase locking method. 8
- b) Determine if the following schedule is serializable. 6
- $S1: r1(X), r2(Z), r1(Z), r3(X), r3(Y), w1(X), w3(Y), r2(X), w2(Z), w1(Y)$
 $S2: r3(X), r2(X), w3(X), r1(X), w1(X).$
- (Note: $ri(X)/wi(X)$ means transaction T_i issues read/write on item X)
- 20 a) Explain column oriented database and document oriented database. 6

- b) Determine if the following schedule is recoverable. Is the schedule cascadeless? 4

Justify your answer.

$r_1(X), r_2(Z), r_1(Z), r_3(X), r_3(Y), w_1(X), c_1, w_3(Y), c_3, r_2(Y), w_2(Z), w_2(Y), c_2$

(Note: $ri(X)/wi(X)$ means transaction T_i issues read/write on item X ; ci

means transaction T_i commits.)

- c) Explain no undo / redo recovery technique in recovery system. 4
